

AICT Project:

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Restaurant Management System:

Code:

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
#include <string.h>
```

```
// --- Helper Functions for Online Compilers ---
```

```
// Clears the screen (Uses "clear" for online Linux servers)
```

```
void clearScreen() {
```

```
    system("clear"); // Note: If you run this locally on Windows later, change "clear" to "cls"
```

```
}
```

```
// Cleans the input buffer to prevent skipping inputs
```

```
void cleanBuffer() {
```

```
    int c;
```

```

    while ((c = getchar()) != '\n' && c != EOF);
}

// Pauses the screen until the user presses Enter
void waitForKey() {
    printf("\n Press Enter to continue...");
    getchar();
}

// --- Original Program Functions ---

char getChoice() {
    char ch;
    printf("\n Enter your choice: ");
    scanf(" %c", &ch);
    cleanBuffer(); // Clean up the newline left by scanf
    return ch;
}

int getQuantity() {
    int qty;
    printf("\n Enter quantity: ");
    while (scanf("%d", &qty) != 1) {
        printf(" Invalid input. Enter a number: ");
        cleanBuffer();
    }
}

```

```

    cleanBuffer(); // Clean up the newline left by scanf
    return qty;
}

char displayMenu(char menu[][30], int size, char start, char end) {
    char choice;
    int valid;
    do {
        clearScreen();
        for(int i = 0; i < size; i++)
            printf("\n %d - %s", i + 1, menu[i]);
        printf("\n");

        choice = getChoice();

        if(choice >= start && choice <= end) {
            valid = 1;
        } else {
            valid = 0;
            printf("\n Invalid choice!");
            waitForKey();
        }
    } while(!valid);
    return choice;
}

```

```
void computeBill(int bill, int orderType) {  
    int discount = 0;  
    int tax = 0;  
    int deliveryCharges = 0;  
    int finalBill = 0;  
  
    clearScreen();  
  
    if(bill >= 8000)  
        discount = bill * 25 / 100;  
    else if(bill >= 5000)  
        discount = bill * 20 / 100;  
    else if(bill >= 3000)  
        discount = bill * 10 / 100;  
  
    tax = bill * 5 / 100;  
  
    if(orderType == 1) {  
        if(bill >= 2000)  
            deliveryCharges = 0;  
        else if(bill >= 1000)  
            deliveryCharges = 50;  
        else  
            deliveryCharges = 100;  
    }  
}
```

```
finalBill = bill - discount + tax + deliveryCharges;
```

```
printf("\n =====");
```

```
if(orderType == 1)
```

```
    printf("\n  ONLINE ORDER - HOME DELIVERY");
```

```
else
```

```
    printf("\n  DINE-IN ORDER");
```

```
printf("\n =====");
```

```
printf("\n Total Bill    : Rs %d", bill);
```

```
printf("\n Discount      : Rs %d", discount);
```

```
printf("\n Tax (5%%)       : Rs %d", tax);
```

```
if(orderType == 1) {
```

```
    printf("\n Delivery Charges: Rs %d", deliveryCharges);
```

```
    if(deliveryCharges == 0)
```

```
        printf(" (FREE)");
```

```
}
```

```
printf("\n -----");
```

```
printf("\n Final Amount    : Rs %d", finalBill);
```

```
printf("\n =====");
```

```
if(orderType == 1) {
```

```
    printf("\n Your order will be delivered in");
```

```
    printf("\n 30-45 minutes to your address.");
```

```
}
```

```
printf("\n Thank you for ordering!");  
}
```

```
int main() {  
    char orderTypeMenu[][30] = {  
        "Dine-In",  
        "Online Order (Home Delivery)",  
        "Exit"  
    };  
    char mainMenu[][30] = {  
        "Burgers", "Pizza", "Rolls", "Sandwich",  
        "BBQ", "Chinese", "Drinks", "Desserts", "Check Out"  
    };  
    char burgerMenu[][30] = {  
        "Spicy Burger", "Tikka Burger", "Plain Burger", "Cheese Burger",  
        "Double Patty", "Mushroom Burger", "Veggie Burger", "Fish Burger", "Go Back"  
    };  
    int burgerPrices[] = {500, 550, 450, 600, 750, 650, 480, 700};  
  
    char pizzaMenu[][30] = {  
        "Large Pizza", "Medium Pizza", "Small Pizza", "Stuffed Crust",  
        "BBQ Pizza", "Pepperoni Pizza", "Veggie Pizza", "Margherita", "Go Back"  
    };  
    int pizzaPrices[] = {1800, 1400, 1000, 2200, 2000, 1900, 1600, 1500};  
  
    char rollMenu[][30] = {
```

```

    "Chicken Roll", "Beef Roll", "Veg Roll", "Zinger Roll",
    "Cheese Roll", "Tikka Roll", "Seekh Kabab Roll", "Egg Roll", "Go Back"
};

int rollPrices[] = {400, 450, 350, 500, 550, 480, 520, 380};

char sandwichMenu[][30] = {
    "Club Sandwich", "Chicken Sandwich", "BBQ Sandwich", "Grilled Sandwich",
    "Veg Sandwich", "Tuna Sandwich", "Egg Sandwich", "Turkey Sandwich", "Go Back"
};

int sandwichPrices[] = {600, 550, 650, 500, 450, 680, 480, 720};

char bbqMenu[][30] = {
    "Chicken Tikka", "Chicken Boti", "Malai Boti", "Beef Kabab",
    "BBQ Platter", "Seekh Kabab", "Reshmi Kabab", "Tandoori Chicken", "Go Back"
};

int bbqPrices[] = {900, 850, 950, 800, 1800, 820, 880, 1200};

char chineseMenu[][30] = {
    "Chicken Chowmein", "Chicken Manchurian", "Sweet & Sour", "Fried Rice",
    "Kung Pao Chicken", "Noodle Soup", "Spring Rolls", "Hot & Sour Soup", "Go Back"
};

int chinesePrices[] = {700, 750, 800, 650, 850, 680, 500, 550};

char drinkMenu[][30] = {
    "Cola", "Orange Juice", "Mint Margarita", "Cold Coffee",
    "Milk Shake", "Lemonade", "Mango Juice", "Green Tea", "Go Back"
}

```

```

};

int drinkPrices[] = {200, 250, 300, 350, 400, 220, 280, 180};

char dessertMenu[][30] = {
    "Chocolate Cake", "Ice Cream", "Brownie", "Custard",
    "Fruit Salad", "Cheesecake", "Pudding", "Tiramisu", "Go Back"
};

int dessertPrices[] = {300, 250, 280, 200, 220, 350, 240, 380};

int bill = 0;

char choice, subChoice, orderTypeChoice;

int checkout = 0;

int quantity = 0;

int price = 0;

int orderType = 0;

char customerName[50];
char customerPhone[15];
char customerAddress[100];

clearScreen();

printf("\n\n");

printf("\n =====");

printf("\n  WELCOME TO KRISPO RESTAURANT");

printf("\n =====");

waitForKey();

```



```
orderTypeChoice = displayMenu(orderTypeMenu, 3, '1', '3');
```

```
if(orderTypeChoice == '3') {  
    clearScreen();  
    printf("\n Thank you for visiting!");  
    waitForKey();  
    return 0;  
}
```

```
if(orderTypeChoice == '1') {  
    orderType = 0;  
    clearScreen();  
    printf("\n =====");  
    printf("\n  DINE-IN ORDER");  
    printf("\n =====");  
    printf("\n You have selected Dine-In.");  
    printf("\n No delivery charges will apply.");  
    waitForKey();  
} else if(orderTypeChoice == '2') {  
    orderType = 1;  
    clearScreen();  
    printf("\n =====");  
    printf("\n  ONLINE ORDER - HOME DELIVERY");  
    printf("\n =====");  
    printf("\n Please provide your details:");
```

```

printf("\n\n Enter your name: ");
fgets(customerName, sizeof(customerName), stdin);
customerName[strcspn(customerName, "\n")] = 0; // Remove newline character


printf("\n Enter your phone: ");
fgets(customerPhone, sizeof(customerPhone), stdin);
customerPhone[strcspn(customerPhone, "\n")] = 0;


printf("\n Enter your address: ");
fgets(customerAddress, sizeof(customerAddress), stdin);
customerAddress[strcspn(customerAddress, "\n")] = 0;


printf("\n -----");
printf("\n Delivery Charges:");
printf("\n Rs 100 (Orders below Rs 1000)");
printf("\n Rs 50 (Orders Rs 1000-1999)");
printf("\n FREE (Orders Rs 2000 & above)");
printf("\n =====");
waitForKey();
}

do {
    choice = displayMenu(mainMenu, 9, '1', '9');
    switch(choice) {
        case '1':

```

```

subChoice = displayMenu(burgerMenu, 9, '1', '9');
if(subChoice >= '1' && subChoice <= '8') {
    price = burgerPrices[subChoice - '1'];
    clearScreen();
    printf("\n =====");
    printf("\n  %s", burgerMenu[subChoice - '1']);
    printf("\n =====");
    printf("\n Price: Rs %d", price);
    quantity = getQuantity();
    if(quantity > 0) {
        bill += price * quantity;
        printf("\n -----");
        printf("\n Item added to cart!");
        printf("\n Quantity: %d", quantity);
        printf("\n Subtotal: Rs %d", price * quantity);
        printf("\n =====");
    }
    waitForKey();
}
break;
case '2':
    subChoice = displayMenu(pizzaMenu, 9, '1', '9');
    if(subChoice >= '1' && subChoice <= '8') {
        price = pizzaPrices[subChoice - '1'];
        clearScreen();
        printf("\n =====");

```

```

printf("\n  %s", pizzaMenu[subChoice - '1']);

printf("\n =====");

printf("\n Price: Rs %d", price);

quantity = getQuantity();

if(quantity > 0) {

    bill += price * quantity;

    printf("\n -----");

    printf("\n Item added to cart!");

    printf("\n Quantity: %d", quantity);

    printf("\n Subtotal: Rs %d", price * quantity);

    printf("\n =====");

}

waitForKey();

}

break;

case '3':

    subChoice = displayMenu(rollMenu, 9, '1', '9');

    if(subChoice >= '1' && subChoice <= '8') {

        price = rollPrices[subChoice - '1'];

        clearScreen();

        printf("\n =====");

        printf("\n  %s", rollMenu[subChoice - '1']);

        printf("\n =====");

        printf("\n Price: Rs %d", price);

        quantity = getQuantity();

        if(quantity > 0) {

```

```

        bill += price * quantity;

        printf("\n -----");

        printf("\n Item added to cart!");

        printf("\n Quantity: %d", quantity);

        printf("\n Subtotal: Rs %d", price * quantity);

        printf("\n =====");

    }

    waitForKey();

}

break;

case '4':

    subChoice = displayMenu(sandwichMenu, 9, '1', '9');

    if(subChoice >= '1' && subChoice <= '8') {

        price = sandwichPrices[subChoice - '1'];

        clearScreen();

        printf("\n =====");

        printf("\n  %s", sandwichMenu[subChoice - '1']);

        printf("\n =====");

        printf("\n Price: Rs %d", price);

        quantity = getQuantity();

        if(quantity > 0) {

            bill += price * quantity;

            printf("\n -----");

            printf("\n Item added to cart!");

            printf("\n Quantity: %d", quantity);

            printf("\n Subtotal: Rs %d", price * quantity);

```

```

        printf("\n =====");
    }
    waitForKey();
}
break;
case '5':
    subChoice = displayMenu(bbqMenu, 9, '1', '9');
    if(subChoice >= '1' && subChoice <= '8') {
        price = bbqPrices[subChoice - '1'];
        clearScreen();
        printf("\n =====");
        printf("\n  %s", bbqMenu[subChoice - '1']);
        printf("\n =====");
        printf("\n Price: Rs %d", price);
        quantity = getQuantity();
        if(quantity > 0) {
            bill += price * quantity;
            printf("\n -----");
            printf("\n Item added to cart!");
            printf("\n Quantity: %d", quantity);
            printf("\n Subtotal: Rs %d", price * quantity);
            printf("\n =====");
        }
        waitForKey();
    }
break;

```

case '6':

```
    subChoice = displayMenu(chineseMenu, 9, '1', '9');  
    if(subChoice >= '1' && subChoice <= '8') {  
        price = chinesePrices[subChoice - '1'];  
        clearScreen();  
        printf("\n =====");  
        printf("\n  %s", chineseMenu[subChoice - '1']);  
        printf("\n =====");  
        printf("\n Price: Rs %d", price);  
        quantity = getQuantity();  
        if(quantity > 0) {  
            bill += price * quantity;  
            printf("\n -----");  
            printf("\n Item added to cart!");  
            printf("\n Quantity: %d", quantity);  
            printf("\n Subtotal: Rs %d", price * quantity);  
            printf("\n =====");  
        }  
        waitForKey();  
    }  
    break;
```

case '7':

```
    subChoice = displayMenu(drinkMenu, 9, '1', '9');  
    if(subChoice >= '1' && subChoice <= '8') {  
        price = drinkPrices[subChoice - '1'];  
        clearScreen();
```

```

printf("\n =====");
printf("\n  %s", drinkMenu[subChoice - '1']);
printf("\n =====");
printf("\n Price: Rs %d", price);
quantity = getQuantity();
if(quantity > 0) {
    bill += price * quantity;
    printf("\n -----");
    printf("\n Item added to cart!");
    printf("\n Quantity: %d", quantity);
    printf("\n Subtotal: Rs %d", price * quantity);
    printf("\n =====");
}
waitForKey();
}

break;

case '8':
    subChoice = displayMenu(dessertMenu, 9, '1', '9');
    if(subChoice >= '1' && subChoice <= '8') {
        price = dessertPrices[subChoice - '1'];
        clearScreen();
        printf("\n =====");
        printf("\n  %s", dessertMenu[subChoice - '1']);
        printf("\n =====");
        printf("\n Price: Rs %d", price);
        quantity = getQuantity();
    }
}

```



```

        if(quantity > 0) {
            bill += price * quantity;

            printf("\n -----");

            printf("\n Item added to cart!");

            printf("\n Quantity: %d", quantity);

            printf("\n Subtotal: Rs %d", price * quantity);

            printf("\n =====");

        }

        waitForKey();

    }

    break;

case '9':

    if(bill > 0) {

        checkout = 1;

    } else {

        clearScreen();

        printf("\n =====");

        printf("\n  Your cart is empty!");

        printf("\n =====");

        printf("\n Please add items before checkout.");

        waitForKey();

    }

    break;

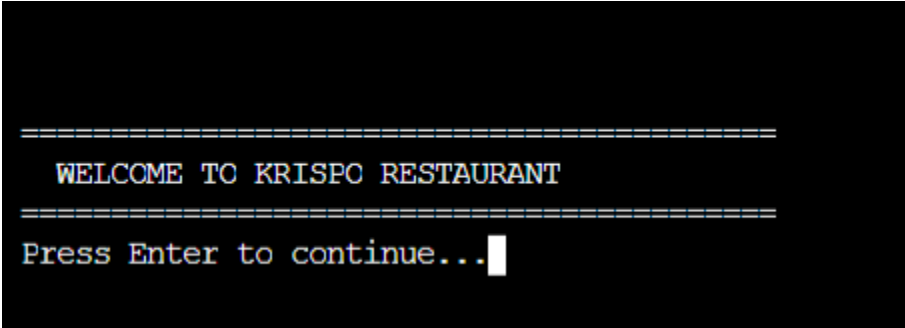
}

} while(!checkout);

```

```
if(orderType == 1) {  
    clearScreen();  
    printf("\n =====");  
    printf("\n  DELIVERY INFORMATION");  
    printf("\n =====");  
    printf("\n Customer: %s", customerName);  
    printf("\n Phone: %s", customerPhone);  
    printf("\n Address: %s", customerAddress);  
    printf("\n =====");  
    waitForKey();  
}  
  
computeBill(bill, orderType);  
waitForKey();  
return 0;  
}
```

Output:



```
=====
WELCOME TO KRISPC RESTAURANT
=====
Press Enter to continue... 
```

- 1 - Dine-In
- 2 - Online Order (Home Delivery)
- 3 - Exit

Enter your choice: 1

=====

DINE-IN ORDER

=====

You have selected Dine-In.
No delivery charges will apply.
Press Enter to continue...

- 1 - Burgers
- 2 - Pizza
- 3 - Rolls
- 4 - Sandwich
- 5 - BBQ
- 6 - Chinese
- 7 - Drinks
- 8 - Desserts
- 9 - Check Out

Enter your choice: 1

- 1 - Spicy Burger
- 2 - Tikka Burger
- 3 - Plain Burger
- 4 - Cheese Burger
- 5 - Double Patty
- 6 - Mushroom Burger
- 7 - Veggie Burger
- 8 - Fish Burger
- 9 - Go Back

Enter your choice: 2

=====

Tikka Burger

=====

Price: Rs 550

Enter quantity: 4

Item added to cart!

Quantity: 4

Subtotal: Rs 2200

=====

Press Enter to continue...

```
1 - Burgers
2 - Pizza
3 - Rolls
4 - Sandwich
5 - BBQ
6 - Chinese
7 - Drinks
8 - Desserts
9 - Check Out
```

```
Enter your choice: 9
```

```
=====
      DINE-IN ORDER
=====
Total Bill      : Rs 2200
Discount        : Rs 0
Tax (5%)        : Rs 110
-----
Final Amount    : Rs 2310
=====
Thank you for ordering!
Press Enter to continue...
```

```
...Program finished with exit code 0
Press ENTER to exit console.
```

Explanation:

This program is a **Restaurant Management System written in C language**. It allows a user to place an order either for **dine-in** or **online home delivery**. The system is designed in a menu-driven format where the user can easily select food items, enter quantity, and generate the final bill.

At the start of the program, the user is welcomed and asked to choose the type of order: dine-in, online order, or exit. If the user selects online order, the program collects basic details such as **name, phone number, and address**. It also shows the delivery charges based on the total bill.

The program contains multiple food categories like **Burgers, Pizza, Rolls, Sandwiches, BBQ, Chinese, Drinks, and Desserts**. Each category has its own submenu with different items and prices. The user can select any item, enter the quantity, and the cost is added to the total bill. This process continues until the user chooses the checkout option.

To make the program user-friendly, helper functions are used such as:

- Clearing the screen
- Taking valid input
- Displaying menus
- Pausing the program until the user presses a key

When the user selects checkout, the program calculates the final bill. It applies:

- **Discounts** based on the total amount (10%, 20%, or 25%)
- **Tax of 5%**
- **Delivery charges** (only for online orders, depending on bill amount)

Finally, the system displays a complete bill including total amount, discount, tax, delivery charges, and final payable amount. For online orders, it also shows a delivery message.
